Register Number: 20101200 7

Name of the Candidate: BALAJI G.

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B.E. DEGREE EXAMINATION, 2024

(COMMON TO CS, IT AND AI & ML)

(FOURTH SEMESTER)

CSBS - 401/ ITBS401/AIBS401- DISCRETE MATHEMATICS

(W.E.F. May 2024)

May] [Time : 3 Hours

Maximum : 75 Marks

PART A - (10 X 1 = 10)

Answer ALL Questions

		Marks	CO	PO	BTL
1.	Define Tautological Implication	1	CO1	1,2	K1
2.	State Absorption laws in logic.	1	CO1	2,3	K2
3.	If A={1} and B={1,2} find A×B	1	CO2	1,2	K1
4.	Define subset of a set. What is mean by proper subset?	1	CO2	1,2	K1
5.	State the modular inequality of a Lattice	1	CO3	2,3	K1
6.	Define Boolean algebra	1	CO3	1,2	K1
7.	Define a Monoid with an example	1	CO4	1,2	K1
8.	What is the identity element of the algebraic system {s,*} Where S is the set of integers and * is defined by $a * b = a + b + 2$ for all $a, b \in S$	1	CO4	3,4	K2
9.	Define regular graph with an example	1	CO5	1,2	K1
10.	Define graph isomorphism.	1	CO5	1,2	K1

PART- B (5 x 3 = 15 Marks)

Answer either (a) or (b) in each Question

11.	(a)	Prove that $P \rightarrow (Q \rightarrow R) \Leftrightarrow (P \wedge Q) \rightarrow R$	3	CO1	1,2	K2
		(OR)				
	(b)	Prove that $\neg Q \wedge (P \rightarrow Q) \Rightarrow \neg P$	3	CO1	2,3	K2
12.	(a)	If A,B and C are any three sets Prove that $A - (B \cup C) = (A - B) \cap (A - C)$	3	CO2	2,3	K2
		(OR)				

(b) If R is the relation on $A = \{1,2,3\}$ such that $(a,b) \in R$ if and only if $a + b =$ even, Find $R^{-1}, \bar{R}$ and R^2 using relational matrix M_R. | 3 | CO2 | 2,3 | K2 |

13. (a) Draw the Hasse diagram for the divisibility relation on the set D_{24} | 3 | CO3 | 1,2 | K2 |

(OR)

(b) State and prove the isotonic property of a Lattice | 3 | CO3 | 2,3 | K2 |

14. (a) Show that in a group G, | 3 | CO4 | 1,2 | K2 |

$(a * b)^{-1} = b^{-1} * a^{-1}$ for all $a, b \in G$

(OR)

(b) Show that $\{(1,2,3,4,5,6), \times_7\}$ is a cyclic group. | 3 | CO4 | 3,4 | K2 |

What are its generators?

15. (a) Explain Konigsberg bridge problem. Represent the problem by means of graph. Does the problem have a solution? | 3 | CO5 | 1,2 | K2 |

(OR)

(b) Define Bipartite graph. Draw $K_{2,3}$ and $K_{3,3}$ graphs. | 3 | CO5 | 2,3 | K2 |

PART- C (4 x10 = 40 Marks)
Answer either (a) or (b) in each Question

16. (a) Obtain the Principal Conjunctive normal form of | 10 | CO1 | 2,3 | K3 |

$(\neg P \rightarrow R) \wedge (Q \rightleftarrows P)$

(OR)

(b) Show that $R \rightarrow S$ can be derived from the premises $P \rightarrow (Q \rightarrow S)$, $\neg R \vee P$ and Q | 10 | CO1 | 3,4 | K3 |

17. (a) Show that the relation R " congruent modulo n" is an equivalence relation on the set of integers | 10 | CO2 | 2,3 | K3 |

(OR)

(b) Prove that $\oplus$ is an associative operation | 10 | CO2 | 1,3 | K3 |

18. (a) Show that in a distributive lattice $\{L, \vee, \wedge\}$ if an element $a \in L$ has a complement then it is unique | 10 | CO3 | 1,2 | K3 |

(OR)

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(b) Simplify the Boolean function $f(a,b,c,d) = \sum(0,1,2,3,4,6,7,8,9,11,15)$ | 10 | CO3 | 2,3 | K3 |

19. (a) Show that $\{s_3,*\}$, where $s = \{1,2,3\}$ is a group under the operation of right composition of permutation | 10 | CO4 | 2,3 | K3 |

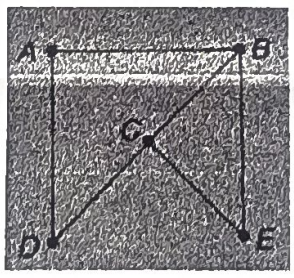
(OR)

(b) List the code words generated by the encoding function $e: B^3 \rightarrow B^6$ whose parity check matrix is | 10 | CO4 | 2,3 | K3 |

$$H = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

PART- D (1 x 10 = 10 Marks)
Answer either (a) or (b)

20. (a) Summarize all the paths of length 4 between D and E for the following graph using adjacency matrix. | 10 | CO5 | 2,3 | K4 |



(OR)

(b) *Evaluate the shortest path matrix of the weighted graph given below using Warshall's algorithm* | 10 | CO5 | 2,3 | K4 |

